

Tensor networks techniques for differential equations: Part II

QTADE School 2026, Bilbao

Stefano Pisoni
Researcher, Quantum Algorithms
QRC, TII Abu Dhabi, UAE
PhD, TUHH

Raghavendra Peddinti
Researcher, Quantum Algorithms
QRC, TII Abu Dhabi, UAE

Outline

Implicit time-stepping

Encoding boundary/initial conditions

Open problems

Recap

Part 1: Function representation using QTTs.



Revisiting of the Heat equation:

$$\partial_t u = \alpha \partial_x^2 u \quad \implies \quad u^{k+1} = (\mathbb{1} + \alpha \Delta t L) u^k$$

- Is Explicit scheme enough?

Motivation

Why do we need implicit solvers?
Stability.

Why do we need stability?
Decouple time and space discretization.

Implicit scheme and stability

$$\partial_t u = \alpha \partial_x^2 u \quad \implies \quad u^{k+1} = (\mathbb{1} + \alpha \Delta t L) u^k$$

Courant criteria: $\alpha \Delta t / \Delta x^2 \leq 1/2$

Implicit Euler time stepping: $(\mathbb{1} - \alpha \Delta t L) u^{k+1} = u^k$

- A-stable w.r.t Δt
- Only accuracy is affected and not stability
- Needs a linear system solver for each timestep

Solving $Au=b$

Direct solvers

Operate on the matrix elements until we have easy to invert matrix

Iterative solvers

Recursively improve the guess until convergence w/ global operations

Variational solvers

Vary the parameters of the solution using a loss function

Solving $Au=b$ within the MPS representation

Some ground rules.

- Assume that the operator A and vector b admit a low-rank MPS.
- We might not know the rank of the solution MPS.
- The ideal solver should be find the solution and its rank.

Useful fact to know:

- MPS of fixed ranks is a manifold.

TT linear system solvers

Direct solvers ✗

No easy access to matrix elements*

Iterative solvers ✓

TT-GMRES, Riemannian solvers

Variational solvers ✓

TT-ALS, TT-mALS, TT-AMEn

* If you're lucky, you can explicitly construct a MPO for the inverse of your operator.

See: Kazeev, V. A., & Khoromskij, B. N. (2012). Low-rank explicit QTT representation of the Laplace operator and its inverse.

Why not iterative methods?

- For fixed-point (CG) or Krylov-based (GMRES) methods:
 - Rank of intermediate steps could blow up despite the final solution having a low rank.
- For Riemannian optimization over the TT manifold:
 - ~~○ We are mortals. And we have hobbies.~~
 - Quite useful for nonlinear problems but for linear problems, unclear if it's worth the extra effort.
 - Needs well-behaved retraction and vector transport routines.
 - Not rank adaptive by construction.

TT-GMRES:

Dolgov, S. V. (2012). TT-GMRES: on solution to a linear system in the structured tensor format. arXiv preprint arXiv:1206.5512.

Riemannian CG:

Kressner, D. et. al. (2016). Preconditioned low-rank Riemannian optimization for linear systems with tensor product structure.

DMRG in a slide.

- Reformulate the linear system as a variational problem:

$$\min_x \|Ax - b\|^2 \quad \text{or} \quad \min_x \frac{1}{2} x^\top Ax - x^\top b \quad (A \succ 0)$$

- Sketch of DMRG:
 - TNs are a multilinear object.
 - So freeze all tensors but one to reduce the problem.
 - Solve the local (i.e. reduced) problem(s) by sweeping through the network.
 - Use gauge freedom to your advantage.

Blackboard.



Variants of DMRG solvers.

TT-ALS

- Single-site updates
- Fixed rank
- Sensitive to initial guess

TT-mALS

- Two-site update
- Rank adaptivity
- Local solves are expensive

TT-AMEn

- Single-site update
- Adaptive rank via subspace enrichment
- Guarantees beyond SPD matrices

TT-ALS and TT-mALS:

Holtz, S., Rohwedder, T., & Schneider, R. (2012). The alternating linear scheme for tensor optimization in the tensor train format.

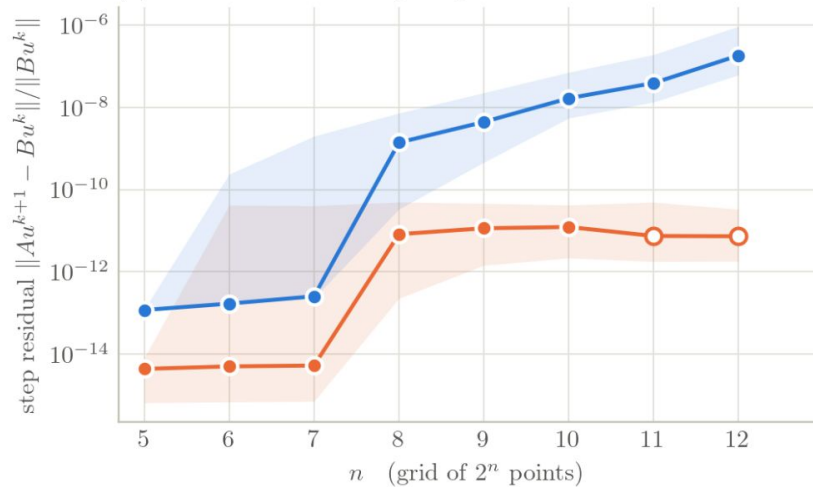
TT-AMEn:

Dolgov, S. V., & Savostyanov, D. V. (2014). Alternating minimal energy methods for linear systems in higher dimensions.

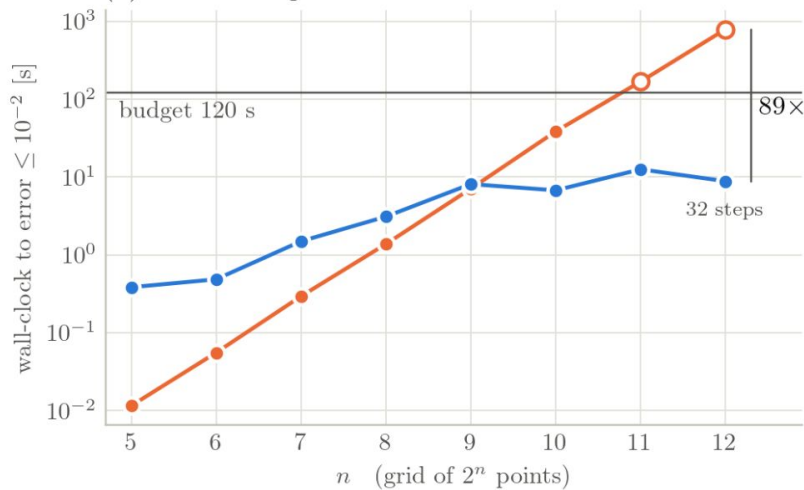
Back to Heat equation.

—●— explicit Euler, $\alpha\Delta t/\Delta x^2 = 0.4$ —●— backward Euler + DMRG, largest Δt meeting ε ○ stopped at 120 s; cost projected

(c) Residual of each step's equation



(d) Time to target error



Which bond dimension? And when to stop?

- Error sources
 - Discretization error (space and time)
 - Representation error (truncation)
 - Solver error (residual)
- Rule of thumb: Ensure discretization error is the dominant error.
- Low residual doesn't mean low error.
 - Use it as certificate of merit instead of certificate of authenticity.
- Real test of a numerical solution:
 - It should converge under grid refinement.

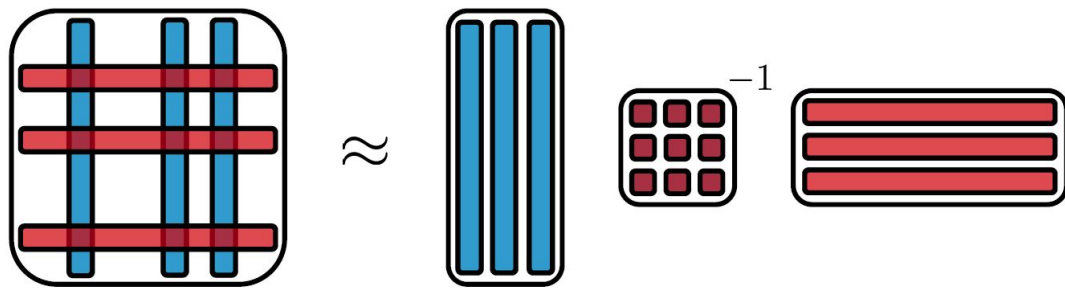
Encoding boundary/initial conditions

- So far we assumed that all the IC/BCs are easily available.
- Now, let's look at how encode arbitrary functions into QTTs.

Encoding functions as QTTs

- Explicit constructions:
 - Polynomials, sine, cosine, exponential
- TT-SVD:
 - Requires function value on the entire grid.
 - Exact and quasi-optimal method.
 - Efficient for reasonable grid sizes.
 - Randomized variants exist.
- TT-Cross (TCI):
 - Queries the function adaptively and interpolates within the QTT.
 - No guarantees of convergence but extremely efficient.

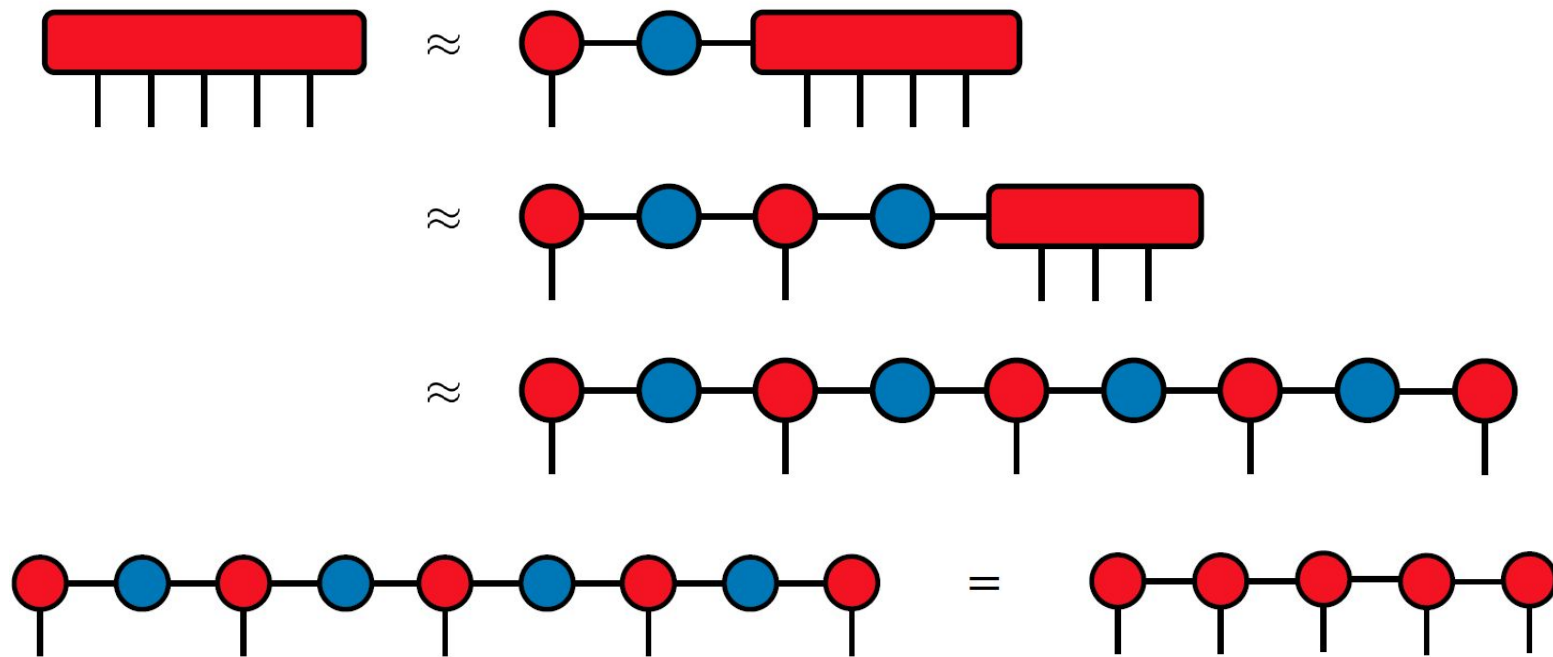
Matrix Cross Interpolation



$$A = A(\mathbb{I}, \mathbb{J}) \approx A(\mathbb{I}, \mathcal{J}) A(\mathcal{I}, \mathcal{J})^{-1} A(\mathcal{I}, \mathbb{J})$$

- Select r columns and rows (also called pivots) and invert the intersecting submatrix (called the Pivot matrix).
- If A is exactly rank r :
 - Heuristic methods find pivots and recover the matrix to machine precision.

Extension to TTs.



Details of TT-Cross.



Variants of TT-Cross.

#01 TCI

Nesting pivots for global interpolation.

#02 Rook search

Greedily increase pivots after each sweep.

#03 Informed pivots

Use structure of the function for initial pivots.

#04 DMRG-Cross

Create 2-site block to search for new pivots.

#05 AMEn-Cross

Use AMEn residual directions for new pivots.

#06 TT-Opt

Using TT-cross for grid-search optimization

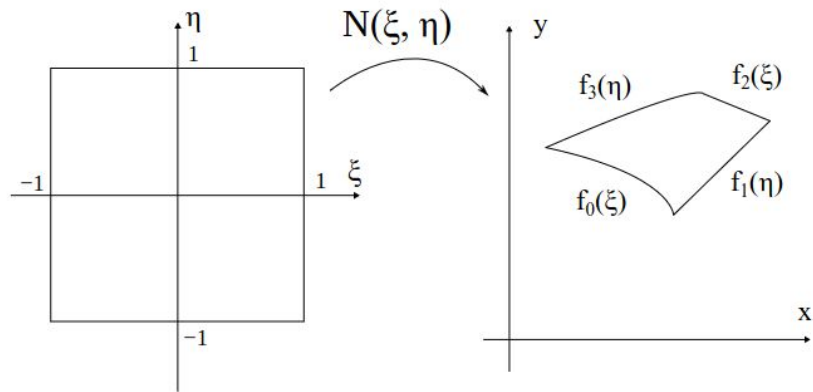
TT-Cross for encoding geometry

Article

Tensor Network Space-Time Spectral Collocation Method for Time-Dependent Convection-Diffusion-Reaction Equations

Dibyendu Adak, Duc P. Truong ^{*}, Gianmarco Manzini , Kim Ø. Rasmussen and Boian S. Alexandrov

$$J = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{bmatrix} = \begin{bmatrix} \frac{\partial N(\xi, \eta)}{\partial \xi} & \frac{\partial N(\xi, \eta)}{\partial \eta} \end{bmatrix}. \quad (7)$$

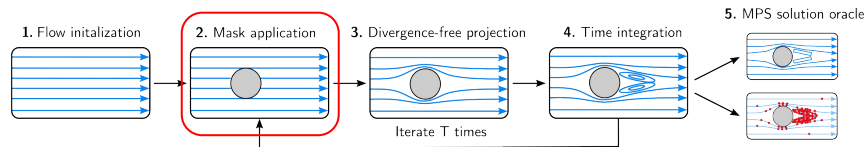
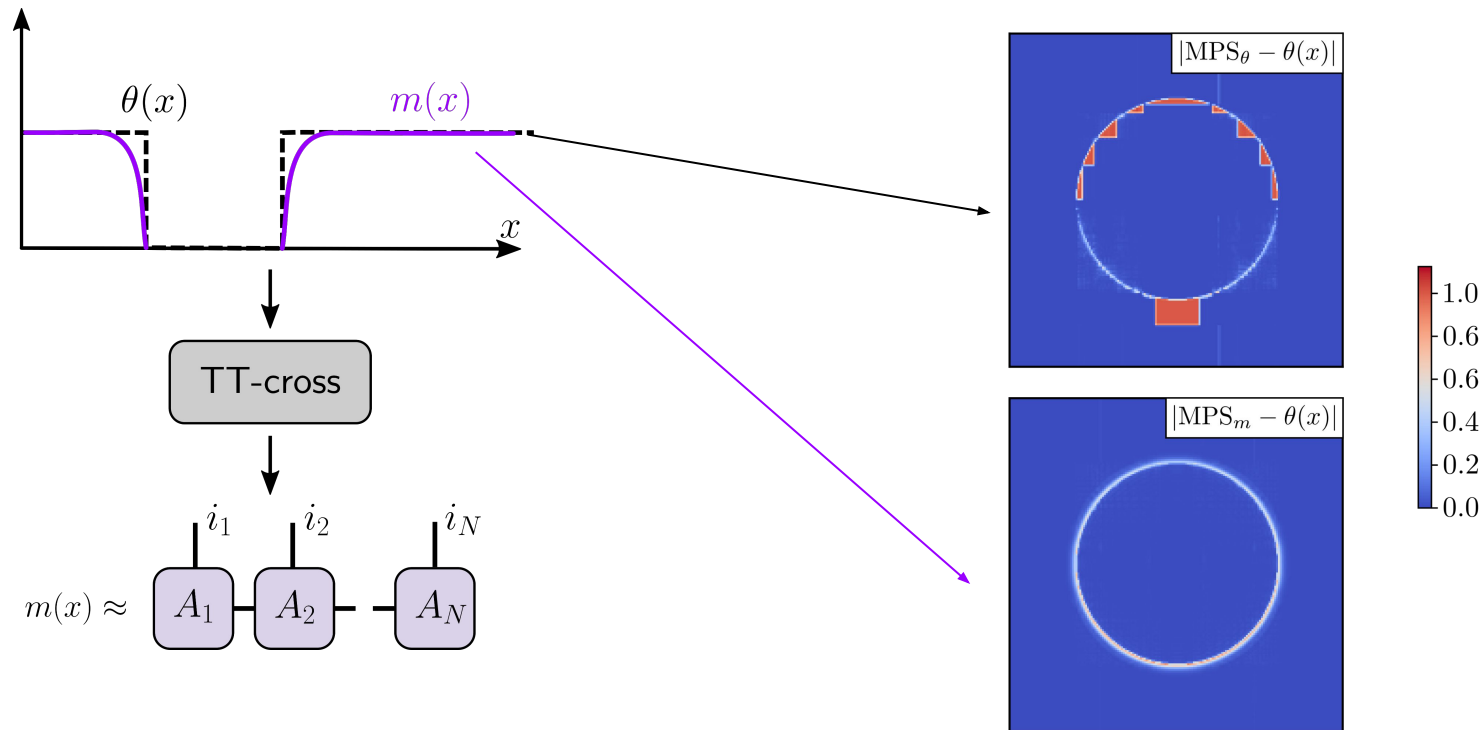


Article

TetraFEM: Numerical Solution of Partial Differential Equations Using Tensor Train Finite Element Method

Egor Kornev ^{1,*}, Sergey Dolgov ^{1,2}, Michael Perelshtein ¹ and Artem Melnikov ¹

MPS mask generation



Interesting questions.

- TT preconditioning for solving stiff problems
 - Bachmayr, M., & Kazeev, V. (2020). Stability of low-rank tensor representations and structured multilevel preconditioning for elliptic PDEs
- Explicit QTT construction using known interpolation schemes
 - Lindsey, M. (2023). Multiscale interpolative construction of quantized tensor trains
 - Guzman, S. E., Tiunov, E., & Aolita, L. (2026). Efficient upsampling for tensor-network and quantum-state encoded functions
- Distributed TT algorithms??
 - Hölscher, L., et. al. (2025). Quantum-inspired fluid simulation of two-dimensional turbulence with GPU acceleration. *Physical Review Research*, 7(1).
 - Röhrig-Zöllner, M., et. al. (2025). Performance of linear solvers in tensor-train format on current multicore architectures
- Robust TT recovery?
 - TT-cross overestimates the ranks under noise. Can this be solved?

Thank you.

- See you tomorrow.

